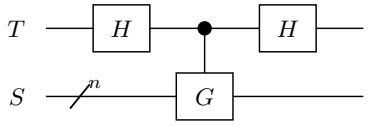
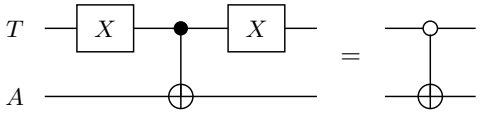
(a) One-bit phase estimation QPE_1 **(`phase_estimation`, one clock qubit)**



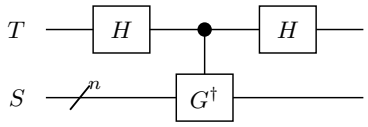
$|0\rangle_T|u_j\rangle \mapsto e^{-i\lambda_j t/2}\big(c_j|0\rangle + i\,s_j|1\rangle\big)_T|u_j\rangle$
$c_j = \cos(\lambda_j t/2)$, $s_j = \sin(\lambda_j t/2)$; the second H is the one-qubit IQFT. Written for e^{-iAt} as in 7.9; the built $G \approx e^{+iAt}$ flips the sign of s_j, which cancels in every probability

(b) Flag F **(`x`, `cx`, `x`)**



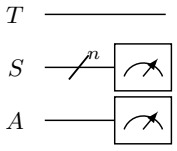
$|0\rangle_T|0\rangle_A \mapsto |0\rangle_T|1\rangle_A$
flag $= 1$ marks the cosine branch

(c) Uncompute $\mathrm{QPE}_1^\dagger$ **(`uncompute_phase_estimation`)**



QFT on one qubit $= H$; c-$G^\dagger$ as in C2c. On each branch of the flag it undoes the mode phase, but because F acted in between the clock is left in $c_j|0\rangle - i\,s_j|1\rangle$ or $s_j|0\rangle + i\,c_j|1\rangle$ (7.9, stage 4)

(d) Readout



not measured (traced out)

$\Pr(\text{flag} = 1) = P = \sum_j \beta_j^2 c_j^2$
$\Pr(i, \text{flag} = 1) = \big[\cos^2(\frac{At}{2})u\big]_i^2 + \big[\frac{1}{2}\sin(At)u\big]_i^2$

The 1BQF is (a), (b), (c), (d) in that order after $H^{\otimes n}$ on S (C7, C8). With $t = \pi/s$, $c_j = 0$ exactly on $\lambda = s$: isolated segments (and isolated hits under $A_1 = sI - A_{\text{hit}}$) never set the flag. Registers: T clock (`time`), S system (`b`), A flag (`ancilla`).